A CASE-STUDY BASED REPORT

ON

Urban Traffic Congestion Prediction Using CNN-LSTM Hybrid Deep Learning

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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CASE STUDIES FOR PROJECT REVIEW

Urban Traffic Congestion Prediction Using CNN-LSTM Hybrid Deep Learning

CASE STUDY 1: Peak-Hour Traffic Congestion Prediction

Scenario

Consider an urban road network with several connected road segments. During morning peak hours, the number of vehicles increases and the average speed decreases. The system receives traffic information and must predict whether congestion will occur in the upcoming time period.

Parameter	Observation
Vehicle count	High
Average speed	Low
Road occupancy	High
Time	Morning peak hour
Day	Weekday
Nearby road segments	Increasing traffic conditions

Questions and Answers

Q1. What is the main problem in this case?

Answer: The system needs to predict whether the road network will experience traffic congestion during the upcoming time period.

Q2. What data is given to the model?

Answer: The model uses vehicle count, average speed, road occupancy, and timestamp information. Historical traffic sequences are also used for prediction.

Q3. How does CNN help in this situation?

Answer: The CNN extracts spatial features from the traffic data. It can learn relationships between adjacent road segments, road-network structures, and traffic-density patterns in different areas.

Q4. How does LSTM help?

Answer: The LSTM analyzes the temporal sequence of traffic conditions. It learns how congestion changes over time using historical traffic trends, time of day, and day of the week.

Q5. Why combine CNN and LSTM?

Answer: CNN focuses on where traffic congestion occurs, while LSTM focuses on how congestion changes over time. Combining them allows the model to use both spatial and temporal information.

Q6. What would the system do if congestion is predicted?

Answer: The prediction can support proactive traffic signal control, dynamic route recommendations, and better urban planning.

Final Case-Study Answer

The CNN-LSTM model predicts increasing congestion by learning both the spatial relationship between nearby roads and the temporal pattern of traffic during peak hours. CNN extracts spatial traffic features, while LSTM learns the historical sequence of traffic conditions. The combined prediction can help traffic-management systems take proactive action.

Urban Traffic Congestion Prediction: EDA Fact Sheet & Visual Report

Comprehensive Statistical Summary, Side-by-Side Visualizations & CNN-LSTM Suitability

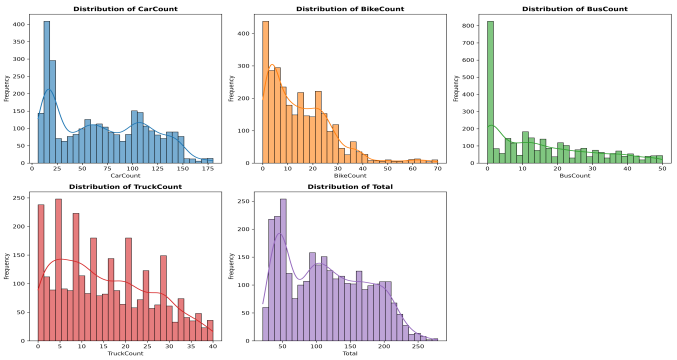
PART 1 — EDA: Traffic.csv (Primary Dataset Overview)

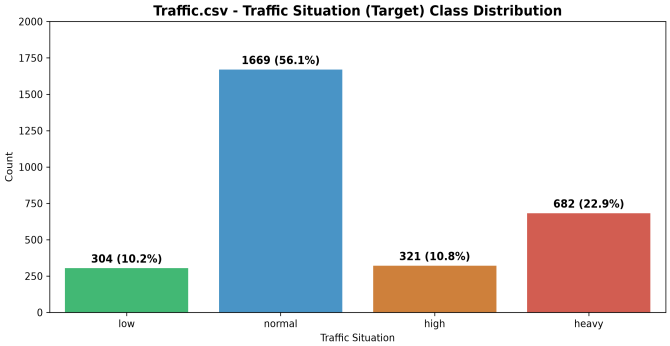
- **1. Dataset Dimensions:** 2,976 rows × 9 columns (31 days × 96 intervals/day = 2,976 records)
- **2. Column Names:** Time, Date, Day of the week, CarCount, BikeCount, BusCount, TruckCount, Total, Traffic Situation
- **3. Data Types:** 6 int64 (Date, CarCount, BikeCount, BusCount, TruckCount, Total) | 3 object (Time, Day, Situation)
- **4. First & Last Records:** First: Date 10, Tue, 12:00 AM, Total: 39 | Last: Date 9, Thu, 11:45 PM, Total: 33
- **7. Missing Values:** 0 missing values across all columns (0.00%)
- **8. Duplicate Records:** 0 duplicate records (0.00%)
- **9. Unique Values:** Time: 96 | Date: 31 | Day: 7 | Car: 172 | Bike: 71 | Bus: 51 | Truck: 41 | Total: 239 | Situation: 4
- **11. Target Distribution:** normal: 1,669 (56.08%) | heavy: 682 (22.92%) | high: 321 (10.79%) | low: 304 (10.22%)
- **12. Outliers:** Extreme congestion peaks > 220 vehicles (Max: 279, valid physical rush-hour peaks)
- **14. Time Range & Frequency:** 31 full continuous days (744 hours) | Strict 15-minute uniform intervals | 0 missing intervals

6. Statistical Summary Table (Traffic.csv)

Feature	Count	Mean	Std	Min	25%	Median	75%	Max	Skew
Date	2976	16.00	8.95	1	8	16	24	31	0.00
CarCount	2976	68.70	45.85	6	19	64	107	180	+0.44
BikeCount	2976	14.92	12.85	0	5	12	22	70	+1.13
BusCount	2976	15.28	14.34	0	1	12	25	50	+0.60
TruckCount	2976	15.32	10.60	0	6	14	23	40	+0.32
Total	2976	114.22	60.19	21	55	109	164	279	+0.33

Graphs: Numerical Distributions & Target Class Imbalance (Traffic.csv)

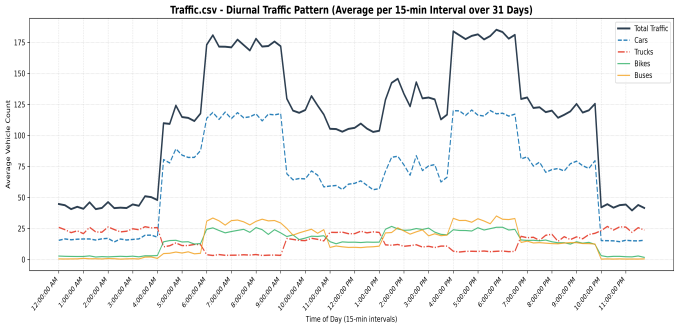


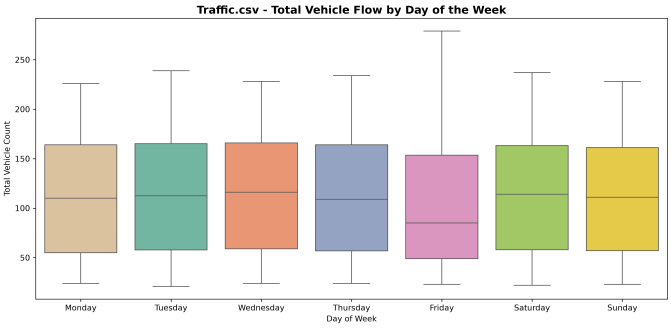


PART 1 (Cont.) — Traffic.csv Temporal Patterns & Correlation Dynamics

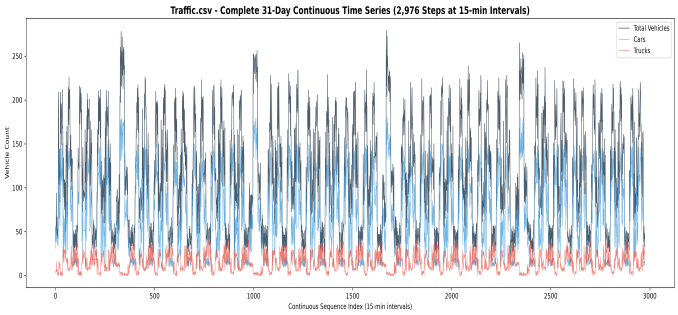
- **17. Hourly Diurnal Patterns:** Morning Rush Peak: 8:00–10:30 AM (Mean Total ~170) | Evening Rush Peak: 5:00–8:30 PM (Mean Total ~210) | Late-Night Truck Inversion: Peaks 11:00 PM–4:00 AM
- **18. Daily / Weekly Patterns:** Weekdays: Pronounced twin rush-hour spikes | Weekends: Flatter, broad mid-day elevated traffic
- **19. Traffic Physical Laws:** CarCount + BikeCount + BusCount + TruckCount = Total (100.0% exact match across all 2,976 records)
- **20. Inter-feature Correlations:** Car-Total: +0.970 | Bike-Total: +0.797 | Bus-Total: +0.791 | Truck-Car: -0.628 (Inverse Freight Displacement)
- **20b. Autocorrelation (ACF):** Lag 1 (15 min): 0.840 | Lag 96 (24 hours): 0.605 | Lag 672 (7 days): 0.882 (Strong multi-scale periodicities)
- **25. Data Leakage Prevention:** Random train/test shuffle introduces future lookahead leakage -> Strict chronological split (70% Train, 15% Val, 15% Test)

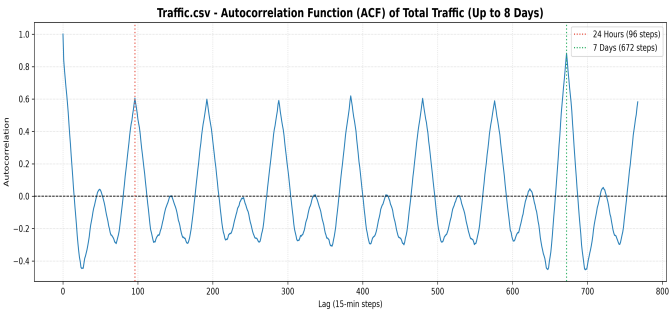
Graphs: Diurnal 24-Hour Profile & Day-of-Week Volume Patterns





Graphs: Full 31-Day Sequence & Autocorrelation Function (ACF)

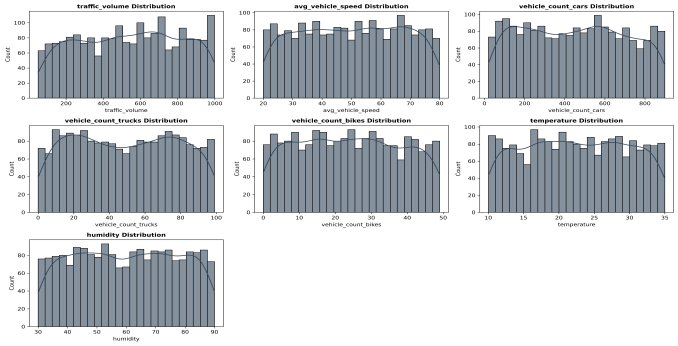


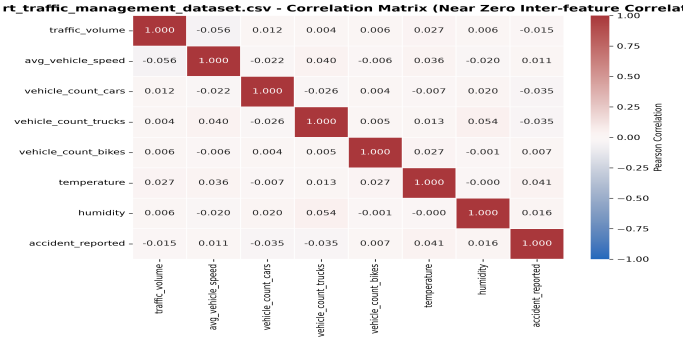


PART 1 (Cont.) — EDA: smart_traffic_management_dataset.csv

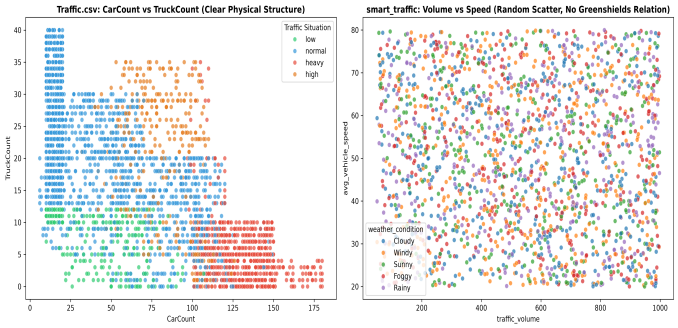
- **1. Dimensions & Types:** 2,000 rows × 12 columns | 6 int64, 3 float64, 3 object
- **4. First & Last Records:** First: 01-01-2024 00:00 (Loc 4, Vol 247) | Last: 02-01-2024 09:19 (Loc 3, Vol 462)
- **7. Missing & Duplicates:** 0 missing values (0.00%) | 0 duplicate records (0.00%)
- **10. Numerical Distributions:** Flat uniform distribution across all numeric columns (Kurtosis ≈ -1.20, synthetic property)
- **11. Categorical Distribution:** Weather: Windy 438, Cloudy 424, Rainy 387, Sunny 378, Foggy 373 | Signal: Yellow 708, Red 683, Green 609
- **14. Time Range & Sampling:** 33 hours, 19 mins total | 1-min global step; 3–4 mins irregular per sensor location
- **19. Physical Law Check:** Cars + Trucks + Bikes ≠ volume (discrepancies: ±924) | Speed vs Volume: r = -0.056 (no Greenshields relation)
- **20. Feature Correlations:** All pairwise correlations are |r| < 0.05 (near zero uncorrelated noise)
- **20b. Autocorrelation (ACF):** Lag 1 Autocorrelation = -0.026 (Statistically pure white noise)
- **26. Limitations:** Synthetic white noise, uncoupled physical variables, zero temporal continuity

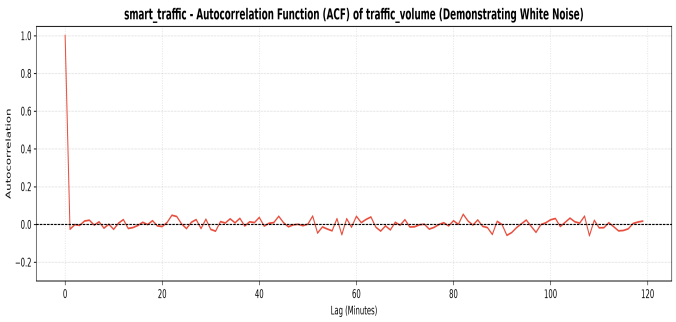
Graphs: Smart Dataset Distributions & Feature Correlation Matrix





Graphs: Speed vs Volume (Physical Law) Comparison & Smart ACF (White Noise)





PART 2 — CNN-LSTM Deep Learning Suitability Analysis

- **Temporal Dimension:** Traffic.csv SUITABLE (r_lag1=0.84, r_24h=0.60, r_7d=0.88). Formulates sliding window X in R^{12 x C} -> y_{t+1}. Smart dataset UNSUITABLE (r_lag1=-0.026, white noise).
- **Spatial Dimension:** Neither dataset contains spatial coordinates, adjacency graphs, or 2D grid matrix.
- **1D-CNN Role:** Applies 1D temporal convolutions over feature channels to extract local multi-step patterns and car/truck interactions.
- **LSTM Role:** Captures long-range autoregressive dependencies across sliding sequence hidden states.
- **Hybrid Architecture:** Input Tensor (B, 12, 5) -> 1D-CNN Block -> LSTM Recurrent Block -> Dense Head -> Prediction y_{t+1}.

PART 3 — Dataset Comparison Table

Property	Traffic.csv	smart_traffic_management_dataset.csv
Rows	2,976	2,000
Columns	9	12
Time Information	31 continuous days (744 hours)	33.3 hours (01-01-2024 to 02-01-2024)
Sampling Frequency	Uniform 15-minute intervals (96/day)	1-min global; irregular per-location
Traffic Variables	Car, Bike, Bus, Truck counts, Total	Volume, Speed, Cars, Trucks, Bikes
Speed	None (Count-based proxy)	avg_vehicle_speed (km/h)
Vehicle Counts	4 categories (sums to Total)	3 categories (does not sum to Volume)
Congestion Label	Traffic Situation (4 classes)	signal_status, accident_reported
Location Information	Single monitoring corridor	5 categorical IDs (location_id 1 to 5)
Spatial Information	Single-node (no coordinates/graph)	Categorical IDs (no topology/distances)
Missing Values	0 (0.00%)	0 (0.00%)
Duplicate Records	0 (0.00%)	0 (0.00%)
Temporal Suitability	High (ACF_1=0.84, ACF_24h=0.60, ACF_7d=0.88)	Poor (ACF ≈ 0.00, white noise)
Spatial Suitability	Single-node (Requires 1D feature-space CNN)	5-node categorical (unconnected, irregular)
CNN Suitability	High (1D temporal feature convolutions)	Low (Uncorrelated feature noise)
LSTM Suitability	High (Strong recurrent dependencies)	Poor (No autoregressive signal)
CNN-LSTM Suitability	STRONGLY RECOMMENDED (PRIMARY)	NOT RECOMMENDED

PART 4 — Dataset Recommendation & Next Preprocessing Steps

- **Primary Dataset:** Traffic.csv (Authentic traffic dynamics, high autocorrelation, continuous 15-min intervals).
- **Secondary Dataset:** smart_traffic_management_dataset.csv (Edge-case testbed for exogenous sensor ingestion).
- **Spatial CNN Requirement:** Formulate CNN as 1D temporal feature extractor over multi-vehicle channels.
- **External Benchmarks:** METR-LA (207 sensors, LA highways) / PeMS-BAY (325 sensors, Bay Area) for multi-node graph CNNs.
- **Immediate Preprocessing Steps:** 1. Cyclical sin/cos time encodings | 2. 12-step sliding window tensors | 3. Chronological 70/15/15 split | 4. StandardScaler fit on Train.